

Interconnection Networks.

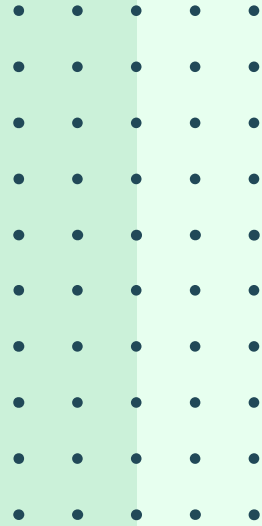
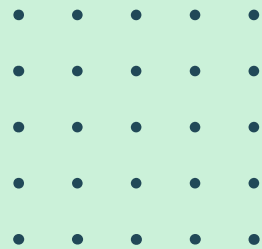


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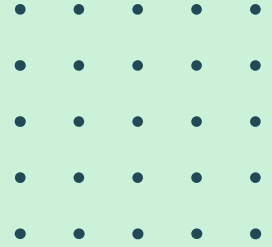
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Learning Objectives



By the end of this presentation, we should be able to:

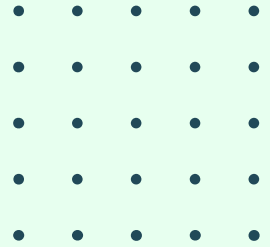
- Define an interconnection network.
- Explain the importance of interconnection networks.
- Identify the basic components of an interconnection network.
- Understand different types of network topologies.
- Compare different topologies based on:
 - Cost
 - Performance
 - Scalability
 - Reliability
- Explain direct and indirect networks.
- Understand switching and routing techniques.

What is an Interconnection Network?

- Connects multiple computing elements.
- Allows components to communicate.
- Transfers data, instructions, and control signals.
- Used in multiprocessor and parallel systems.
- Can connect processors, memory, and I/O devices.

Simple idea:

Source → Network → Destination



Why Are Interconnection Networks Needed?

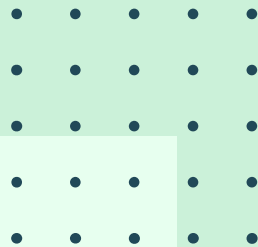
- Modern computers use multiple processors.
- Processors need to communicate with each other.
- Shared memory requires fast data access.
- Large systems need scalable communication.
- Reduces communication bottlenecks.
- Improves overall system performance.

A poorly designed network can become a **bottleneck**.

A good network provides:

- High speed
- Low latency
- High bandwidth
- Reliability
- Scalability

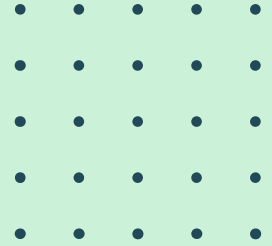
Application Areas



Interconnection networks are used in:

- Multiprocessor systems
- Supercomputers
- Data centers
- Parallel computing
- Distributed systems
- High-performance computing
- Network-on-Chip systems

Network Topology

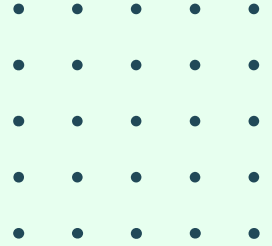


What is Network Topology?

- **Network topology** refers to the arrangement of:
 - Nodes
 - Links
 - Communication paths
- It describes how devices are connected to each other.
- Topology affects:
 - Network performance
 - Cost
 - Reliability
 - Scalability
 - Fault tolerance
- Different applications require different topologies.

Common topologies:

- Bus Topology
- Crossbar Topology
- Mesh Topology
- Torus Topology
- Tree Topology

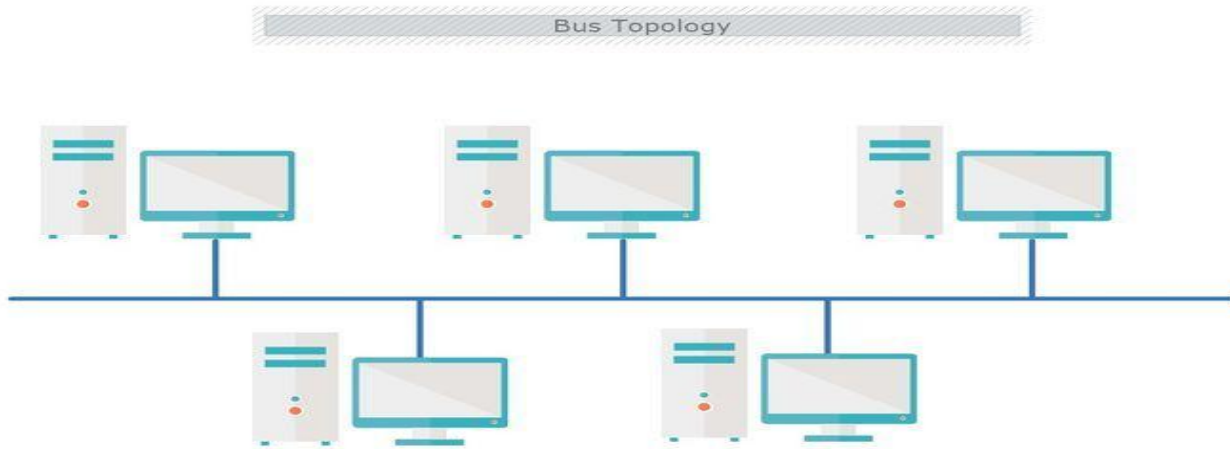
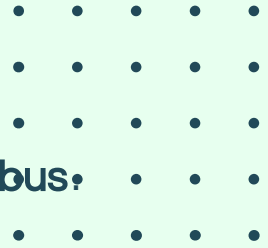


They differ in:

- Cost
- Speed
- Scalability
- Complexity
- Fault tolerance

Bus Topology

- All devices are connected to a single shared communication line called a **bus**.
- Data travels through the common communication medium.
- Only the appropriate device accepts the transmitted data.

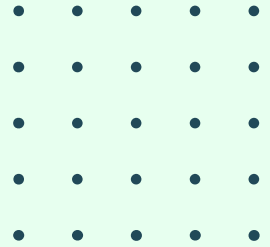


Advantages

- Simple design.
- Low installation cost.
- Requires less cabling.
- Easy to implement for small systems.

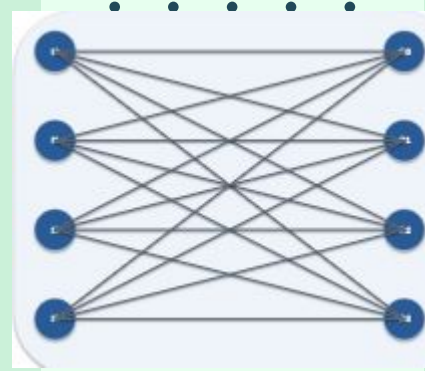
Disadvantages

- Shared bandwidth.
- Collisions/contention can occur.
- Performance decreases as the number of devices increases.
- Failure of the main bus can affect the whole network.
- Not suitable for large systems.



Crossbar Topology

- Uses switches to connect inputs and outputs.
- Provides direct connections between devices.
- Multiple transfers can occur simultaneously.
- High communication performance.
- Common in high-performance systems.



Crossbar Working

- Each intersection contains a switching element.
- Switch establishes a connection between source and destination.
- Multiple connections can operate simultaneously.
- Avoids a single shared communication path.

Example:

CPU1 ————— Memory2
CPU2 ————— Memory1
CPU3 ————— Memory3

All can communicate at the same time

Advantages & Disadvantages of Crossbar



Advantages

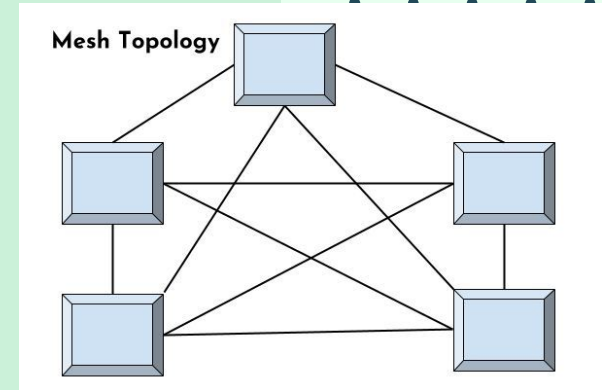
- Very fast communication
- High bandwidth
- Multiple simultaneous transfers
- Low contention

Disadvantages

- Expensive
- Requires many switches
- Hardware complexity increases rapidly
- Difficult to scale for large systems

Mesh Topology

- Devices are arranged in rows and columns.
- Each node connects to nearby nodes.
- Interior nodes usually have four neighbors.
- Common in parallel computers and NoCs.
- Easy to expand.



Characteristics :

- Two-dimensional arrangement.
- Each node has a unique position.
- Data travels through neighboring nodes.
- Routing is relatively simple.
- More scalable than a bus.

Example: A packet from:

Node A → Node B may travel:

A → → → B

or

A

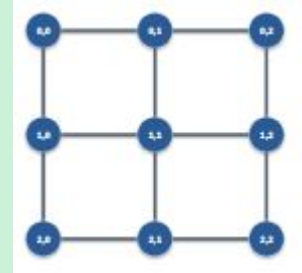
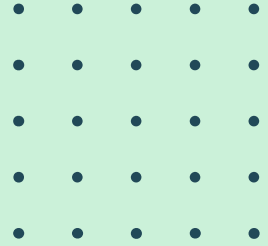
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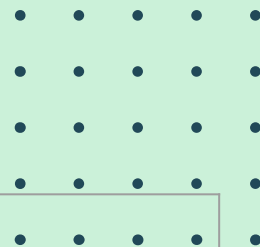
B

Torus Topology

- Extension of mesh topology.
- Opposite edges are connected.
- Provides wrap-around connections.
- Each node can have up to four neighbors.
- Reduces distance between nodes.



Mesh vs Torus



Mesh Topology	Torus Topology
No wrap-around	Wrap-around connections
Edge nodes have fewer links	More uniform connections
Longer paths possible	Shorter paths
Simpler	More complex
Easier to implement	Better connectivity

Key point:

Torus = Mesh + Wrap-around connections

Tree Topology

- Hierarchical network structure.
- Nodes are arranged in levels.
- Root node is at the top.
- Parent nodes connect to child nodes.
- Useful for hierarchical communication.

Tree Characteristics

- Hierarchical structure.
- Easy to organize.
- Data can move upward or downward.
- Suitable for large structured systems.
- Can support many nodes.

Examples:

- Processor hierarchies
- File systems
- Network structures
- Distributed systems

Advantages

- Easy to manage
- Structured communication
- Easy to expand
- Suitable for hierarchical systems

Disadvantages

- Root can become a bottleneck.
- Failure of higher-level nodes affects many nodes.
- Communication may require multiple hops.
- Performance depends on tree structure.

What is Routing?

- Routing determines how data moves through a network.
- A message is divided into packets.
- Packets travel from source to destination.
- Routers/switches select the path.

Basic process:

Source



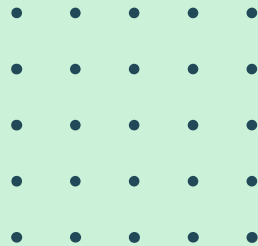
Router



Router



Destination



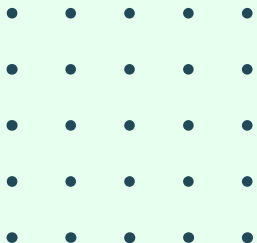
Routing in Interconnection Networks

Routing decides:

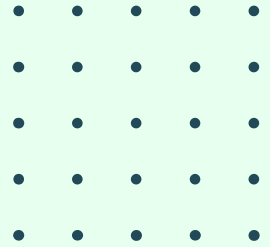
- Source node
- Destination node
- Path to follow
- Direction of movement
- How to avoid blocked paths

Main goals:

- Reduce latency
- Avoid congestion
- Improve bandwidth utilization
- Provide reliable communication



Routing Methods



1. Deterministic Routing

- Always selects the same path.
- Simple and predictable.

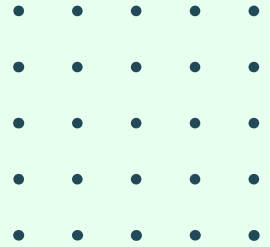
2. Adaptive Routing

- Path changes depending on network conditions.
- Can avoid congestion.

3. Minimal Routing

- Uses the shortest possible path.
- Reduces number of hops.

Example: XY Routing



Used commonly in 2D mesh networks.

- First move in the **X direction** .
- Then move in the **Y direction** .
- Simple and deterministic.

Example:

Source



Move X → → →



↓ Y



Destination

Benefit:

- Simple
- Predictable
- Easy to implement

Performance & NoC

Network Performance

The performance of an interconnection network is commonly measured using:

- Latency
- Bandwidth
- Bisection Width

These measurements tell us how quickly and efficiently data can move.

Latency

- Latency = Time taken for data to reach the destination
- Measures communication delay.
- Lower latency means faster communication.
- Includes transmission and routing delays.
- Important for real-time and parallel systems.

Example:

If a packet takes **5 ns** to reach its destination:

Latency = 5 ns

Bandwidth

Bandwidth = Amount of data transferred per unit time

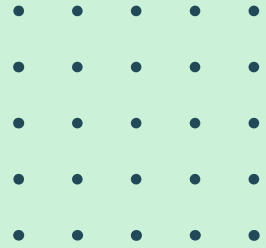
Usually measured in:

- bits/second
- Mbps
- Gbps
- Tbps

Example: A network transferring: **10 GB/s** has higher bandwidth than one transferring:

2 GB/s Higher bandwidth → More data can be transferred.

Bisection Width



- Measures the communication capacity of a network.
- Divide the network into two equal halves.
- Count the minimum number of links crossing the division.
- More crossing links generally mean higher communication capacity.

Comparison of Performance Measures

Measure	Meaining	Desired
Latency	Time to deliver data	Low
Bandwidth	Data transferred per second	High
Bisection width	Links across network division	High

Important: Low latency + High bandwidth + High bisection width = Better network

Network-on-Chip (NoC)

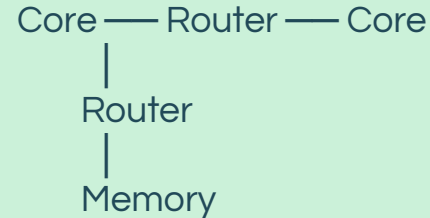
What is NoC?

- NoC = **Network-on-Chip**
- Communication system inside a processor chip.
- Connects processing cores and other components.
- Uses network concepts for on-chip communication.
- Designed for many-core processors.

Advantages of NoC

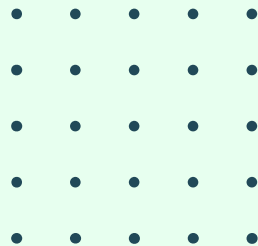
- Scalable
- Supports many processing cores
- Better communication organization
- Reduces communication bottlenecks
- Can provide high bandwidth

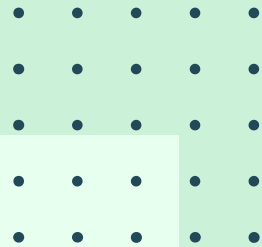
Basic structure:



Conclusion:

- Interconnection networks enable communication.
- Different topologies have different benefits.
- Routing controls data movement.
- NoC brings networking concepts inside modern chips.





Thankyou